



Networkforyou

Subscribe to our
You Tube Channel



Networkforyou



**Welcome
To
Network for you
AAA**



Networkforyou



Authentication
Who are you?

Authorization
How much can you spend?

Accounting
What did you spend it on?

Statement of Personal Credit Card Account

Account Number: 1234-567-890 | Statement Closing Date: 01-31-01 | Current Amount Due: \$278.50

JOE EMPLOYEE
456 SKYVIEW DRIVE
HOMETOWN, USA 99900-1234

MAIL PAYMENT TO:
THE BANK
132 VINE STREET
ANYTOWN, USA 87500-0010

872919345 00178255000000003

Detach here and return upper portion with check or money order. Do not staple or fold.

Statement of Personal Credit Card Account
Retain this portion for your files.

Cardmember Name: JOE EMPLOYEE | Account Number: 1234-567-890 | Statement Closing Date: 01-31-01

Statement Date: 02-01-01 | Payment Due Date: 03-01-01

Closing Date: 01-31-01

Credit Limit: \$1,500.00 | Credit Available: \$1221.50

New Balance: \$278.50 | Minimum Payment Due: \$20.00

Account Summary

Previous Balance:	+74.24	Transaction Fees:	+3.00
Purchases:	+250.50	Annual Fees:	+25.00
Cash Advances:	+0	Current Amount Due:	+250.50
Payments:	-74.25	Amount Past Due:	+0
Finance Charge:	+0	Amount Over Credit Line:	+0
Late Charge:	+0	NEW BALANCE:	\$278.50

Reference Number	Sold	Posted	Activity Since Last Statement	Amount
43210987	01-03	01-13	Payment, Thank You	-\$74.25
01234567	01-12	01-13	Wings 'N' Things Anytown, USA	\$25.25
78901234	01-14	01-17	Record Release Anytown, USA	\$40.00
45678901	01-14	01-17	Sports Stadium Anytown, USA	\$75.25
3210987	01-22	01-23	Tie Tack Anytown, USA	\$20.75
76543210	01-29	01-30	Electronic World Anytown, USA	\$89.25
2345678		01-30	Transaction Fees	\$3.00
34567890		01-01	Annual Fee	\$25.00

AAA (Authentication, Authorization and Accounting):

- Authentication: Verify the identity of the user, who are you.
- Authorization: What is the user allowed to do? Example what resource he can access etc? (How much you can spend)
- Accounting: It is like all record what is done by that user it will keep all record. Example used for billing and auditing (What did you spend it on record)

Email us:
networkforyou4@gmail.com

2 of 7

WhatsApp Us : **+918143809578**



AAA Stand for Authentication, Authorization and Accounting:

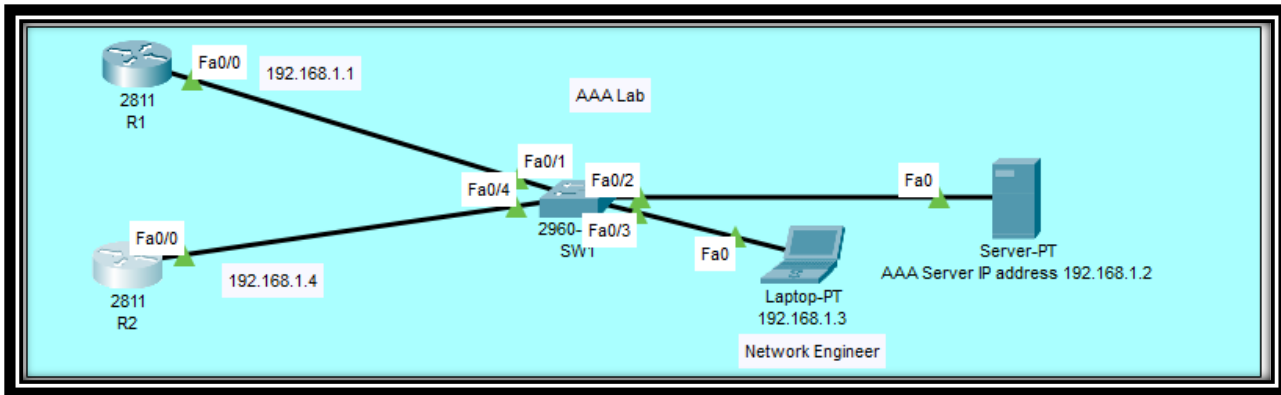
- It is a Centralized Management of users to access the network devices
- AAA Server allow setting up access control on Cisco Routers and Switches
- Like if we have 300 Switches and 10 Router in our organization then it will be very difficult to create all user in that all devices and delete when they leave organization etc. And it will take lot of memory of device also to overcome with this type of issues we use AAA Server.
- AAA Server also control connections passing through router or switch for access network.
- When every user tries to connect to router or switch these network devices verifies by AAA Server (AAA Database)
- User Management is done with AAA Server without need to reconfigure to individual router or switch
- Like any new user came we need to configure only in AAA Server no need to add that user in Router or switch.
- AAA Server use two Main type of Protocol to configure this
- Radius Protocol (Remote Authentication Dial-in User Service)
- TACACS+ (Terminal Access Controller Access-Control System Plus)

Radius Protocol:

- It is open standard where as TACACS is Cisco Proprietary protocol.
- It uses UDP and users ports numbers 1812/1645 and 1813/1646.
- It Encrypts passwords only.
- It is light weight protocol (Consume less resources).

TACACS+:

- It is CISCO Proprietary protocol.
- It use TCP and port number user 49.
- It encrypts entire communication.
- It is heavy weight protocol consuming more resources.



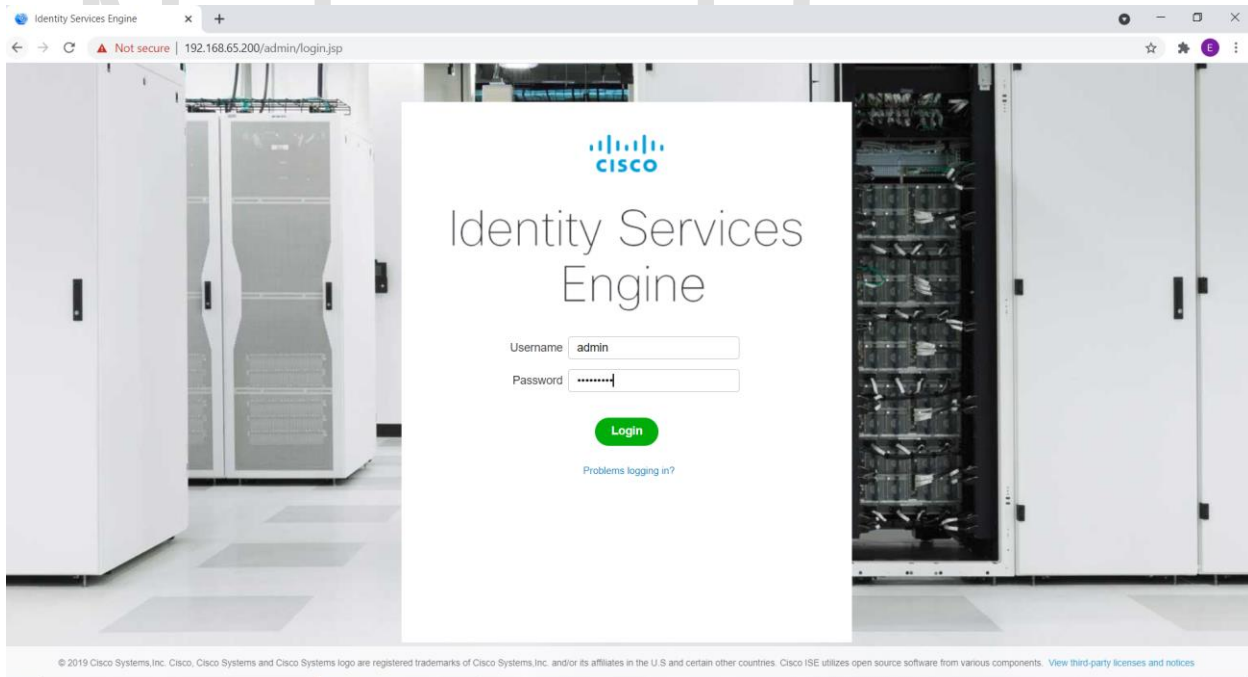
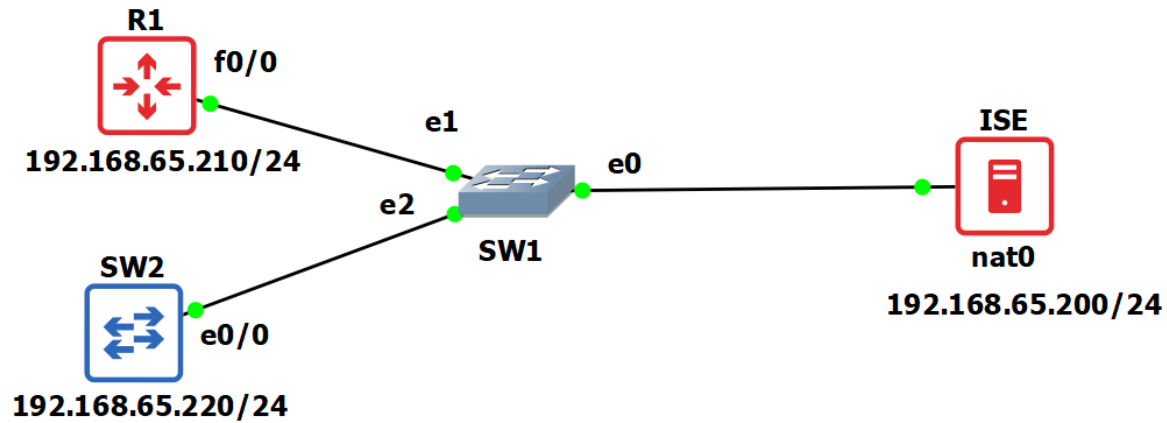
R1 Configuration	R2 Configuration
<pre>en config t hostname R1 int f0/0 ip add 192.168.1.1 255.255.255.0 no sh aaa new-model Tacacs-server host 192.168.1.2 key abc123 aaa authentication login AAA group tacacs+ Line vty 0 1 login authentication AAA</pre>	<pre>en config t hostname R2 int f0/0 ip add 192.168.1.4 255.255.255.0 no sh aaa new-model radius-server host 192.168.1.2 key abc123 aaa authentication login AAA group radius Line vty 0 1 login authentication AAA</pre>

AAA with ISE Lab:

Email us:
networkforyou4@gmail.com

4 of 7

WhatsApp Us : +918143809578



Email us:
networkforyou4@gmail.com

5 of 7

WhatsApp Us : +918143809578



Networkforyou

Subscribe to our
You Tube Channel

Login to Cisco ISE and Register devices

The screenshot shows the Cisco Identity Services Engine (ISE) interface. The top navigation bar includes Home, Context Visibility, Operations, Policy, Administration, and Work Centers. The left sidebar shows Network Devices, Default Device, and Device Security Settings. The main content area displays a table of Network Devices with columns for Name, IP/Mask, Profile Name, Location, Type, and Description. Two devices are listed: R1 (Router) and SW1 (Switch).

Name	IP/Mask	Profile Name	Location	Type	Description
R1	192.168.65.210/32	Cisco	All Locations	All Device Types	Router
SW1	192.168.65.220/32	Cisco	All Locations	All Device Types	Switch

Go to Administrator then Deployment then enable device admin service

The screenshot shows the Cisco Identity Services Engine (ISE) Deployment page. The left sidebar shows Deployment, Licensing, Certificates, Logging, Maintenance, Upgrade, Backup & Restore, Admin Access, and Settings. The main content area displays the deployment configuration for the ISE1 node. The Role is set to STANDALONE. The Administration checkbox is checked. The Monitoring checkbox is checked, and the Role is set to PRIMARY. The Policy Service checkbox is checked, and the Enable Session Services checkbox is checked. The Enable Device Admin Service checkbox is checked and highlighted with a red box. The Save button is visible at the bottom.

Go to Administration then Deployment then enable device admin service as show below

Email us:
networkforyou4@gmail.com

6 of 7

WhatsApp Us : +918143809578



Networkforyou

Subscribe to our
You Tube Channel

Networkforyou

Email us:
networkforyou4@gmail.com

7 of 7

WhatsApp Us : +918143809578